OnlineBid

An Online Auctioning System

Software Requirements Specification (Functional Requirements)

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1. Introduction

1.1 Purpose

This document specifies the functional requirements of **OnlineBid**, a web-based auctioning platform. It describes what the system must do from the point of view of its users so that it can be designed, implemented and tested against a clear, agreed baseline.

1.2 Scope

OnlineBid allows registered sellers to list items for auction and registered buyers to place competitive bids on those items in real time. The system manages user accounts, auction listings, bidding, winner selection, payment recording, notifications and administration. Physical delivery of goods and the processing of money by banks or mobile-money providers are outside the system; OnlineBid integrates with external payment gateways only to confirm payments.

1.3 Definitions and Abbreviations

Term	Meaning
Auction	A time-limited sale in which an item is sold to the highest bidder.
Bid	An offer of money placed by a buyer on an active auction.
Reserve price	The minimum price a seller will accept; if not reached, the item is not sold.
Starting price	The lowest amount at which bidding may begin.
Bid increment	The minimum amount by which a new bid must exceed the current highest bid.
Proxy bid	A maximum amount a buyer sets so the system bids automatically on their behalf.
SRS	Software Requirements Specification.

1.4 Intended Audience

Lecturers, project supervisors, developers, testers and any stakeholder who needs to understand the behaviour of the system.

2. Overall Description

2.1 User Classes

User class	Description
Guest	Unregistered visitor who can browse and search auctions but cannot bid or sell.
Buyer (Bidder)	Registered user who places bids, watches auctions and pays for won items.
Seller	Registered user who creates auction listings and manages sales.
Administrator	Staff member who manages users, listings, disputes and system settings.

2.2 Assumptions and Dependencies

- Users have internet access and a modern web browser or mobile device.
- An external payment gateway and an email/SMS service are available.

- All auction times are handled in a single server time zone and shown in the user's local time.

3. Functional Requirements

Requirements are identified as FR-x.y. Priority is **High** (essential), **Medium** (important) or **Low** (desirable). The word *shall* marks a mandatory requirement.

3.1 User Registration and Authentication

ID	Requirement	Priority
FR-1.1	The system shall allow a guest to register using a name, email address, phone number and password.	High
FR-1.2	The system shall verify a new account through a confirmation link or one-time code sent to the user's email or phone.	High
FR-1.3	The system shall allow registered users to log in and log out securely.	High
FR-1.4	The system shall allow users to reset a forgotten password.	High
FR-1.5	The system shall allow users to view and update their profile (contact details, address, profile photo).	Medium
FR-1.6	The system shall allow a registered user to act as both buyer and seller using one account.	Medium

3.2 Auction Listing Management (Seller)

ID	Requirement	Priority
FR-2.1	The system shall allow a seller to create an auction by entering a title, description, category, condition, starting price, bid increment, start time and end time.	High
FR-2.2	The system shall allow a seller to upload one or more images of the item.	High
FR-2.3	The system shall allow a seller to set an optional reserve price and an optional 'Buy It Now' price.	Medium
FR-2.4	The system shall allow a seller to edit or cancel an auction only before the first bid is placed.	High
FR-2.5	The system shall allow a seller to schedule an auction to start at a future date and time.	Medium
FR-2.6	The system shall allow a seller to view all their auctions and their status (scheduled, active, ended, sold, unsold).	High
FR-2.7	The system shall allow a seller to relist an unsold item.	Low

3.3 Browsing and Searching

ID	Requirement	Priority
FR-3.1	The system shall display a list of active auctions to all users, including guests.	High
FR-3.2	The system shall allow users to search auctions by keyword.	High
FR-3.3	The system shall allow users to filter auctions by category, price range, condition and time remaining.	High
FR-3.4	The system shall allow users to sort auctions by price, time remaining and newest.	Medium

ID	Requirement	Priority
FR-3.5	The system shall display an auction detail page showing images, description, current highest bid, number of bids, time remaining, seller information and bid history.	High

3.4 Bidding

ID	Requirement	Priority
FR-4.1	The system shall allow a logged-in buyer to place a bid on an active auction.	High
FR-4.2	The system shall reject a bid that is lower than the starting price or lower than the current highest bid plus the bid increment, and shall show an explanatory message.	High
FR-4.3	The system shall prevent a seller from bidding on their own auction.	High
FR-4.4	The system shall update the current highest bid and the bid history immediately after each valid bid, without requiring a manual page refresh.	High
FR-4.5	The system shall allow a buyer to set a maximum (proxy) bid and shall bid automatically on their behalf up to that amount.	Medium
FR-4.6	The system shall allow a buyer to purchase an item instantly at the 'Buy It Now' price, ending the auction, where that option is available.	Medium
FR-4.7	The system shall extend the auction end time by a fixed period (e.g. 2 minutes) if a bid is placed in the last minutes of the auction (anti-sniping).	Low
FR-4.8	The system shall not allow bids on auctions that have ended or been cancelled.	High

3.5 Auction Closing and Winner Selection

ID	Requirement	Priority
FR-5.1	The system shall automatically close an auction when its end time is reached.	High
FR-5.2	The system shall declare the highest bidder the winner if the reserve price (when set) has been met; otherwise the auction shall be marked as unsold.	High
FR-5.3	The system shall notify the winner, the seller and, where applicable, outbid users of the result.	High
FR-5.4	The system shall record the final price and the winner against the auction.	High

3.6 Payment and Transaction Management

ID	Requirement	Priority
FR-6.1	The system shall allow the winning buyer to pay for the item through an integrated payment method (e.g. mobile money or card).	High
FR-6.2	The system shall record each transaction with its amount, date, payer, payee and payment status.	High
FR-6.3	The system shall notify the seller once payment has been confirmed so that the item can be shipped or handed over.	High
FR-6.4	The system shall allow the buyer to mark an item as received and shall then mark the transaction as complete.	Medium
FR-6.5	The system shall allow users to view their transaction history and download receipts.	Medium

ID	Requirement	Priority
FR-6.6	The system shall flag a winning buyer who does not pay within a set period and notify the seller and administrator.	Medium

3.7 Watchlist and Notifications

ID	Requirement	Priority
FR-7.1	The system shall allow a buyer to add auctions to a personal watchlist.	Medium
FR-7.2	The system shall notify a buyer by email, SMS or in-app message when they are outbid.	High
FR-7.3	The system shall notify watchers when a watched auction is about to end (e.g. one hour before).	Medium
FR-7.4	The system shall allow users to choose which notifications they receive.	Low

3.8 Ratings, Reviews and Communication

ID	Requirement	Priority
FR-8.1	The system shall allow a buyer and seller to rate each other and leave a review after a completed transaction.	Medium
FR-8.2	The system shall display a seller's average rating and reviews on their profile and listings.	Medium
FR-8.3	The system shall allow buyers to send questions to a seller about an item, and allow the seller to reply.	Low

3.9 Administration

ID	Requirement	Priority
FR-9.1	The system shall allow an administrator to view, suspend, reactivate or delete user accounts.	High
FR-9.2	The system shall allow an administrator to review, edit or remove auction listings that violate platform rules.	High
FR-9.3	The system shall allow an administrator to manage item categories.	Medium
FR-9.4	The system shall allow an administrator to view and resolve disputes between buyers and sellers.	Medium
FR-9.5	The system shall allow an administrator to generate reports on users, auctions, bids and revenue (e.g. commission fees).	Medium
FR-9.6	The system shall allow an administrator to configure platform settings such as commission rate and minimum bid increment.	Low

4. Use Case Summary

Actor	Use cases
Guest	Register, browse auctions, search and filter auctions, view auction details

Actor	Use cases
Buyer	Log in, place bid, set proxy bid, buy now, manage watchlist, pay for item, confirm receipt, rate seller
Seller	Create auction, upload images, edit or cancel auction, view bids, relist item, rate buyer, answer questions
Administrator	Manage users, manage listings and categories, resolve disputes, generate reports, configure settings

4.1 Main Scenario: Placing a Bid

Item	Description
Actor	Buyer
Precondition	The buyer is logged in and the auction is active.
Main flow	1. Buyer opens the auction detail page. 2. Buyer enters a bid amount. 3. System validates the amount against the current highest bid and increment. 4. System records the bid and updates the current price. 5. System notifies the previous highest bidder that they were outbid.
Alternative flow	If the amount is too low or the auction has ended, the system rejects the bid and displays an error message.
Postcondition	The new bid is stored and the buyer becomes the current highest bidder.

5. Supporting Non-Functional Requirements (Brief)

Although this document focuses on functional requirements, the following qualities support them:

- **Performance:** new bids shall appear to other viewers within 2 seconds.
- **Security:** passwords shall be stored hashed and all data transmitted over HTTPS.
- **Reliability:** auctions shall close at the exact scheduled time, and no valid bid shall be lost.
- **Usability:** the interface shall be responsive and usable on mobile phones.
- **Availability:** the system shall target 99% uptime.